



6384 - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar System

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Inner Main Belt Families				
	1	2441	NIRSpec IFU Spectroscopy	(1) 2441
	2	6712	NIRSpec IFU Spectroscopy	(2) 6712
	3	6769	NIRSpec IFU Spectroscopy	(3) 6769
	103	6769	NIRSpec IFU Spectroscopy	(3) 6769
	5	495	NIRSpec IFU Spectroscopy	(4) 495
	6	163	NIRSpec IFU Spectroscopy	(5) 163
	7	752	NIRSpec IFU Spectroscopy	(6) 752
	8	302	NIRSpec IFU Spectroscopy	(7) 302

Folder	Observation	Label	Observing Template	Science Target
	9	313	NIRSpec IFU Spectroscopy	(9) 313
	109	313	NIRSpec IFU Spectroscopy	(9) 313
	10	84	NIRSpec IFU Spectroscopy	(8) 84

ABSTRACT

Collisional impacts between asteroid-sized objects are common in our solar system history and have played a significant role in forming asteroid families [1]. Low-albedo (geometric albedo < 0.07) and low-inclination ($i < 15^\circ$) asteroid families in the inner main belt (IMB) between the ν_6 secular resonance at ~ 2.15 AU and the 3:1 mean-motion resonance with Jupiter at ~ 2.5 AU are particularly intriguing because they were proposed to be the possible sources of low-albedo and water-rich (hereafter primitive) near-Earth asteroids (NEAs) [2, 3]. Primitive NEAs are the parent bodies of the least-altered meteorites, the carbonaceous chondrites, and harbor valuable historical records of the early solar system [4]. These NEAs are rich in water and carbonaceous materials, critical key elements relevant to the origin of life on Earth. Investigating the origin and source of primitive NEAs in the IMB will help us trace the origin of water and complex organics that may have been delivered to Earth by these NEAs. Here, we propose to use JWST's NIRSpec Integral Field Unit (IFU) to acquire spectra ($0.6\text{--}5.3\ \mu\text{m}$) of IMB family members and directly compare them to those of primitive NEAs to test the dynamical theories that postulate that the IMB region is the source of primitive NEAs.

OBSERVING DESCRIPTION

Our low-albedo and inclination inner main belt family targets include members of the New Polana, Eulalia, Erigone, Sulamitis, Clarissa, Chaldaea, and Klio families. We propose using NIRSpec Integral Field Unit (IFU) to acquire our targets' low-resolution spectra ($\sim 0.6\text{--}5.3\ \mu\text{m}$) and investigate their surface composition. The instrument setup will include the NRSIRS2RAPID readout pattern with a 4-POINT-DITHER to reduce overheads. All objects will be observable by NIRSpec for over a month during the Cycle 3 observational window. These objects have well-defined orbits, which have negligible uncertainty in the ephemerides, $> 0.15''$. The object's V magnitudes were estimated by the JPL Horizons tools and vary between 13.84 and 19.66, in their faintest appearance. Additionally, all objects are outside of the galactic plane and are not affected by field stars' contamination in Cycle 3. For the brightest objects in our target list, we will use the medium-resolution gratings (G140M/F100LP, G235M/F170LP, G395M/F290LP), while for the faintest we will use the PRISM mode. Exposure times were estimated in a way that our targets can be observed at any observation window in Cycle 3 without risk of saturation, and reaching the desired SNR to address our proposed science goals.

Proposal 6384 - Targets - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar System

Solar System Targets	#	Name	Level 1	Level 2	Level 3
	(1)	2441	TYPE=ASTEROID,A=2.408796089561816,E=0.1931 154284699811,I=3.74231584529955 ,O=140.1358103072589,W=217.0352124021415,M=2 47.4712895807911,EQUINOX=J2000,EPOCH=22- MAR-2016:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=Unknown				
	(2)	6712	TYPE=ASTEROID,A=2.39889162101958,E=0.12339 80991802157,I=0.968644506687826 ,O=264.1674391993938,W=322.3566053271125,M=3 56.2915539121113,EQUINOX=J2000,EPOCH=18- SEP-2016:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=Unknown				
	(3)	6769	TYPE=ASTEROID,A=2.416276793133673,E=0.1232 880596041379,I=3.899266189115711 ,O=59.00035442281334,W=55.01933337829828,M=2 34.4283463333609,EQUINOX=J2000,EPOCH=01- MAY-2017:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=Unknown				
	(4)	495	TYPE=ASTEROID,A=2.487142199984987,E=0.1297 335593745075,I=2.279484256172301 ,O=186.4792385919962,W=206.9562390179051,M=2 7.38424494970353,EQUINOX=J2000,EPOCH=26- OCT-2016:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=Unknown				
	(5)	163	TYPE=ASTEROID,A=2.366608427802293,E=0.1911 111285246242,I=4.813850271283584 ,O=160.1981199657965,W=298.3687644083784,M=2 91.0652635233349,EQUINOX=J2000,EPOCH=19- JAN-2013:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=Unknown				
	(6)	752	TYPE=ASTEROID,A=2.463006791146524,E=0.0734 8076330528111,I=5.959305017491489 ,O=85.17159450824154,W=23.44235455730605,M=1 88.2702234513804,EQUINOX=J2000,EPOCH=07- OCT-2014:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=Unknown				
	(7)	302	TYPE=ASTEROID,A=2.406491143895606,E=0.1114 15118679767,I=3.413121710900026 ,O=7.861581612274328,W=54.87545084782993,M=7 6.9348286590799,EQUINOX=J2000,EPOCH=02- MAR-2015:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=Unknown				
	(8)	84	TYPE=ASTEROID,A=2.361810548127401,E=0.2365 668301355955,I=9.331147660147447 ,O=327.5976287055954,W=14.6441087884156,M=18. 67374781737911,EQUINOX=J2000,EPOCH=15- OCT-2014:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=Unknown				
	(9)	313	TYPE=ASTEROID,A=2.375040983007197,E=0.1808 521227692197,I=11.65355478091188 ,O=176.6381042335611,W=316.0708490107803,M=7. 237554852381169,EQUINOX=J2000,EPOCH=25- AUG-2017:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=Unknown				

Proposal 6384 - Observation 1 - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar Sy...

Observation	Proposal 6384, Observation 1: 2441											Fri Jan 23 22:00:17 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (2441 (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1			Level 2			Level 3			
	(1)	2441	TYPE=ASTEROID,A=2.408796089561816,E=0.1931 154284699811,I=3.74231584529955 ,O=140.1358103072589,W=217.0352124021415,M=2 47.4712895807911,EQUINOX=J2000,EPOCH=22- MAR-2016:00:00:00,EpochTimeScale=TDB									
	Comments: Extended=Unknown											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	2	1	false	true	NONE	4	4	175.067	37
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 2441 FROM JWST LESS THAN 0.075											

Proposal 6384 - Observation 2 - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar Sy...

Observation	Proposal 6384, Observation 2: 6712											Fri Jan 23 22:00:17 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(6712 (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(2)	6712	TYPE=ASTEROID,A=2.39889162101958,E=0.12339 80991802157,I=0.968644506687826 ,O=264.1674391993938,W=322.3566053271125,M=3 56.2915539121113,EQUINOX=J2000,EPOCH=18- SEP-2016:00:00:00,EpochTimeScale=TDB									
	Comments: Extended=Unknown											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	12	1	false	true	NONE	4	4	758.622	38
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 6712 FROM JWST LESS THAN 0.075											

Proposal 6384 - Observation 3 - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar Sy...

Observation	Proposal 6384, Observation 3: 6769												Fri Jan 23 22:00:17 GMT 2026
	Diagnostic Status: Warning												
Observing Template: NIRSpec IFU Spectroscopy													
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(6769 (Obs 3)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(3)	6769	TYPE=ASTEROID,A=2.416276793133673,E=0.1232 880596041379,I=3.899266189115711 ,O=59.00035442281334,W=55.01933337829828,M=2 34.4283463333609,EQUINOX=J2000,EPOCH=01-MAY-2017:00:00:00,EpochTimeScale=TDB										
Comments: Extended=Unknown													
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	PRISM/CLEAR	NRSIRS2RAPID	3	1	false	true	NONE	4	4	233.422	39	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 6769 FROM JWST LESS THAN 0.075												

Proposal 6384 - Observation 103 - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar ...

Observation	Proposal 6384, Observation 103: 6769												Fri Jan 23 22:00:17 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 103:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (6769 (Obs 103)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(3)	6769	TYPE=ASTEROID,A=2.416276793133673,E=0.1232 880596041379,I=3.899266189115711 ,O=59.00035442281334,W=55.01933337829828,M=2 34.4283463333609,EQUINOX=J2000,EPOCH=01-MAY-2017:00:00:00,EpochTimeScale=TDB Comments: Extended=Unknown										
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	PRISM/CLEAR	NRSIRS2RAPID	3	1	false	true	NONE	4	4	233.422	39	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 6769 FROM JWST LESS THAN 0.075												

Proposal 6384 - Observation 5 - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar Sy...

Observation	Proposal 6384, Observation 5: 495 Fri Jan 23 22:00:17 GMT 2026											
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (495 (Obs 5)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2			Level 3		
	(4)	495	TYPE=ASTEROID,A=2.487142199984987,E=0.1297 335593745075,I=2.279484256172301 ,O=186.4792385919962,W=206.9562390179051,M=2 7.38424494970353,EQUINOX=J2000,EPOCH=26- OCT-2016:00:00:00,EpochTimeScale=TDB <i>Comments: Extended=Unknown</i>									
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140M/F100LP	NRSIRS2RAPID	17	1	false	true	NONE	4	4	1050.4	1
	2	G235M/F170LP	NRSIRS2RAPID	10	1	false	true	NONE	4	4	641.911	2
	3	G395M/F290LP	NRSIRS2RAPID	11	1	false	true	NONE	4	4	700.267	3
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 495 FROM JWST LESS THAN 0.075											

Proposal 6384 - Observation 6 - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar Sy...

Observation	Proposal 6384, Observation 6: 163												Fri Jan 23 22:00:17 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(163 (Obs 6)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(5)	163	TYPE=ASTEROID,A=2.366608427802293,E=0.1911111285246242,I=4.813850271283584,O=160.1981199657965,W=298.3687644083784,M=291.0652635233349,EQUINOX=J2000,EPOCH=19-JAN-2013:00:00:00,EpochTimeScale=TDB										
	Comments: Extended=Unknown												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F100LP	NRSIRS2RAPID	8	1	false	true	NONE	4	4	525.2	4	
	2	G235M/F170LP	NRSIRS2RAPID	5	1	false	true	NONE	4	4	350.133	5	
	3	G395M/F290LP	NRSIRS2RAPID	4	1	false	true	NONE	4	4	291.778	6	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 163 FROM JWST LESS THAN 0.075												

Proposal 6384 - Observation 7 - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar Sy...

Observation	Proposal 6384, Observation 7: 752												Fri Jan 23 22:00:17 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(752 (Obs 7)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(6)	752	TYPE=ASTEROID,A=2.463006791146524,E=0.0734 8076330528111,I=5.959305017491489 ,O=85.17159450824154,W=23.44235455730605,M=1 88.2702234513804,EQUINOX=J2000,EPOCH=07- OCT-2014:00:00:00,EpochTimeScale=TDB										
	Comments: Extended=Unknown												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F100LP	NRSIRS2RAPID	11	1	false	true	NONE	4	4	700.267	10	
	2	G235M/F170LP	NRSIRS2RAPID	7	1	false	true	NONE	4	4	466.844	11	
	3	G395M/F290LP	NRSIRS2RAPID	4	1	false	true	NONE	4	4	291.778	12	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 752 FROM JWST LESS THAN 0.075												

Proposal 6384 - Observation 8 - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar Sy...

Observation	Proposal 6384, Observation 8: 302												Fri Jan 23 22:00:17 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (302 (Obs 8)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(7)	302	TYPE=ASTEROID,A=2.406491143895606,E=0.1114 15118679767,I=3.413121710900026 ,O=7.861581612274328,W=54.87545084782993,M=7 6.9348286590799,EQUINOX=J2000,EPOCH=02- MAR-2015:00:00:00,EpochTimeScale=TDB Comments: Extended=Unknown										
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F100LP	NRSIRS2RAPID	11	1	false	true	NONE	4	4	700.267	16	
	2	G235M/F170LP	NRSIRS2RAPID	7	1	false	true	NONE	4	4	466.844	17	
	3	G395M/F290LP	NRSIRS2RAPID	9	1	false	true	NONE	4	4	583.556	18	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 302 FROM JWST LESS THAN 0.075												

Proposal 6384 - Observation 9 - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar Sy...

Observation	Proposal 6384, Observation 9: 313												Fri Jan 23 22:00:17 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(313 (Obs 9)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(9)	313	TYPE=ASTEROID,A=2.375040983007197,E=0.1808 521227692197,I=11.65355478091188 ,O=176.6381042335611,W=316.0708490107803,M=7. 237554852381169,EQUINOX=J2000,EPOCH=25- AUG-2017:00:00:00,EpochTimeScale=TDB										
	Comments: Extended=Unknown												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F100LP	NRSIRS2RAPID	4	1	false	true	NONE	4	4	291.778	25	
	2	G235M/F170LP	NRSIRS2RAPID	3	1	false	true	NONE	4	4	233.422	26	
	3	G395M/F290LP	NRSIRS2RAPID	2	1	false	true	NONE	4	4	175.067	27	
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 313 FROM JWST LESS THAN 0.075												

Proposal 6384 - Observation 109 - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar ...

Observation	Proposal 6384, Observation 109: 313											Fri Jan 23 22:00:17 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 109:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(313 (Obs 109)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(9)	313	TYPE=ASTEROID,A=2.375040983007197,E=0.1808 521227692197,I=11.65355478091188 ,O=176.6381042335611,W=316.0708490107803,M=7. 237554852381169,EQUINOX=J2000,EPOCH=25- AUG-2017:00:00:00,EpochTimeScale=TDB									
	Comments: Extended=Unknown											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size	Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140M/F100LP	NRSIRS2RAPID	4	1	false	true	NONE	4	4	291.778	25
	2	G235M/F170LP	NRSIRS2RAPID	3	1	false	true	NONE	4	4	233.422	26
	3	G395M/F290LP	NRSIRS2RAPID	2	1	false	true	NONE	4	4	175.067	27
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 313 FROM JWST LESS THAN 0.075											

Proposal 6384 - Observation 10 - Low-Albedo and Inclination Asteroid Families as Tracers for Water and Organics in the Inner Solar S...

Observation	Proposal 6384, Observation 10: 84 Fri Jan 23 22:00:17 GMT 2026											
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (84 (Obs 10)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2			Level 3		
	(8)	84	TYPE=ASTEROID,A=2.361810548127401,E=0.2365 668301355955,I=9.331147660147447 ,O=327.5976287055954,W=14.6441087884156,M=18. 67374781737911,EQUINOX=J2000,EPOCH=15- OCT-2014:00:00:00,EpochTimeScale=TDB <i>Comments: Extended=Unknown</i>									
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140M/F100LP	NRSIRS2RAPID	3	1	false	true	NONE	4	4	233.422	25
	2	G235M/F170LP	NRSIRS2RAPID	3	1	false	true	NONE	4	4	233.422	26
	3	G395M/F290LP	NRSIRS2RAPID	3	1	false	true	NONE	4	4	233.422	27
Special Requirements	DEFAULT WINDOW: ANGULAR RATE 84 FROM JWST LESS THAN 0.075											